

LAPORAN PRAKTIKUM 4
SISTEM BASIS DATA



Disusun Oleh :

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Kelas Praktikum/Semester :

B/III

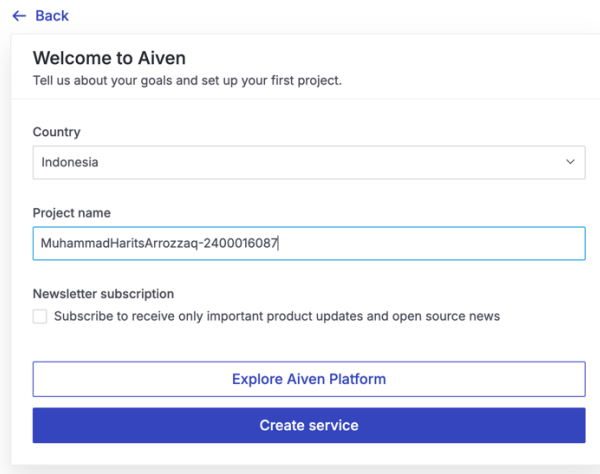
PROGRAM STUDI SISTEM INFORMASI
FAKULTAS SAINS & TEKNOLOGI TERAPAN
UNIVERSITAS AHMAD DAHLAN
YOGYAKARTA
2025

A. Dasar Teori

Aiven adalah layanan *Database as a Service (DBaaS)* berbasis *cloud* yang memudahkan pengguna dalam membuat, mengelola, dan mengakses database tanpa perlu instalasi server lokal. Dalam praktikum Sistem Basis Data, Aiven digunakan untuk membangun dan mengelola database MySQL secara *remote*, sehingga mahasiswa dapat memahami penerapan konsep basis data modern yang terintegrasi dengan layanan *cloud*. Penggunaan Aiven juga membantu menjaga keamanan, ketersediaan, dan kemudahan pengelolaan data secara efisien.

B. Langkah – Langkah Percobaan

1. Create an Aiven Account



← Back

Welcome to Aiven
Tell us about your goals and set up your first project.

Country
Indonesia

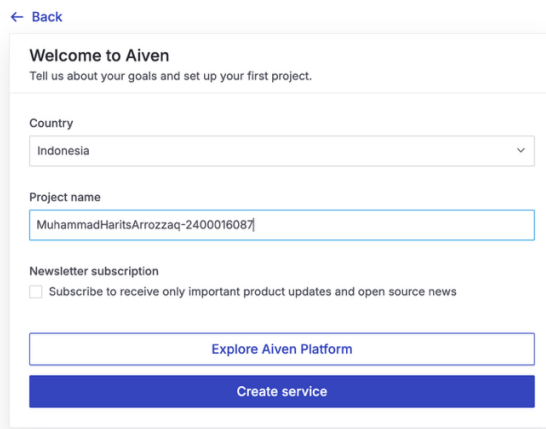
Project name
MuhammadHaritsArrozzaq-2400016087

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Explore Aiven Platform

Create service

2. Create a New Project



← Back

Welcome to Aiven
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Country
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Explore Aiven Platform

Create service

3. Build MySQL Service

Create service
Project: muhammadharitsarrozzaq-2400016087 Organization: My Organization

[Back](#)

Service tier

Free

- One free managed MySQL service
- Good for building a small application and learning to use cloud databases
- No credit card required

Professional

From \$19 trial

- 30-day trial with \$300 of free usage included
- Good for proof of concepts and testing. Open for production use cases
- No credit card required

Region

You can select a specific cloud provider and region on the "Professional" tier.

Asia Pacific **Australia** **Europe** **North America**

Plan	VMs	CPUs per VM	RAM per VM	Storage	Monthly price
Free-1-1gb	1	1	1 GB	1 GB	Free

Service basics

Name*
The service name cannot be changed afterwards

sbd-87

[Add tag to this service](#)

Service summary

Service: MySQL

Name: sbd-87

Service tier: Free

Region: Asia Pacific

Plan: Free-1-1gb
@ 1 CPU @ 1 GB RAM @ 1 GB storage
@ Backups for disaster recovery

Monthly price: **Free**

[Create service](#)

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My Organization / muhammadharitsarrozzaq-2400016087 / sbd-87 / Overview

sbd-87
MySQL 8 [Help](#) [Docs](#) [Notes](#)

Get started with Aiven for MySQL*
Set up, secure, and integrate your Aiven service with a step-by-step guide.
[Get started](#)

Connection information [Quick connect](#)

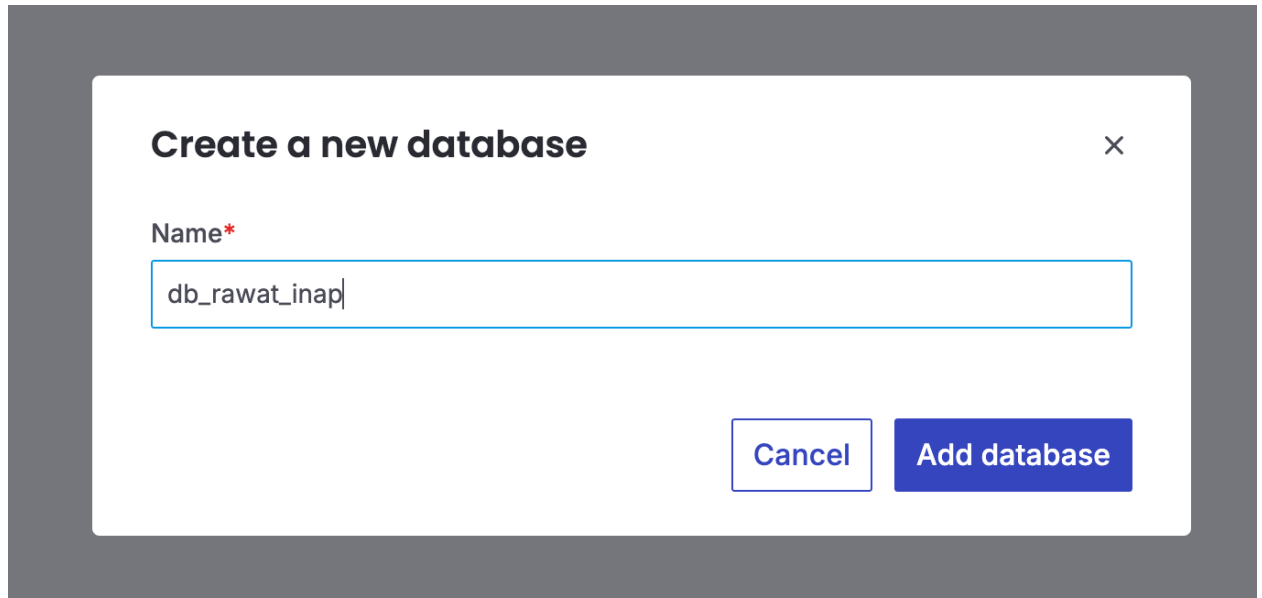
MySQL	MySQLx
Service URI	mysql://CLICK_TO_REVEAL_PASSWORD@sbd-87-muhammadharitsarrozzaq-2400016087.c.aivencloud.com:23744/defaultdb?ssl-mode=REQUIRED
Database name	defaultdb
Host	sbd-87-muhammadharitsarrozzaq-2400016087.c.aivencloud.com
Port	23744
User	avnadmin
Password	*****
SSL mode	REQUIRED
CA certificate	Show

4. Access the Database with Remote Tools



C. Langkah – Langkah Praktikum

1. Create a new database named **db_rawat_inap**



Create a new database ×

Name*

db_rawat_inap

Cancel Add database

2. Create tables:

Patients (Pasien)

```
3 CREATE TABLE Patients (  
4     patient_id INT(10) PRIMARY KEY AUTO_INCREMENT,  
5     name VARCHAR(100) NOT NULL,  
6     gender ENUM('Male', 'Female') NOT NULL,  
7     age INT(10),  
8     address VARCHAR(100),  
9     phone_number VARCHAR(15)  
10 );
```

```
12 INSERT INTO Patients (name, gender, age, address, phone_number) VALUES  
13 ('Bukhari', 'Male', 22, 'Jl. Umbulharjo No. 99, Yogyakarta', '081122334455'),  
14 ('Ariffandi', 'Male', 20, 'Jl. Magelangan No. 35, Wates', '089876543210'),  
15 ('Wowiling', 'Female', 33, 'Jl. Buntu No. 10, Yogyakarta', '08192837465');
```

	patient_id	name	gender	age	address	phone_number	
	1	Bukhari	Male	22	Jl. Umbulharjo No. 99, Yogyakarta	081122334455	
	2	Ariffandi	Male	20	Jl. Magelangan No. 35, Wates	089876543210	
	3	Wowiling	Female	33	Jl. Buntu No. 10, Yogyakarta	08192837465	
		NULL	NULL	NULL	NULL	NULL	

Rooms (Kamar)

```
17 • CREATE TABLE Rooms (  
18     room_id INT(10) PRIMARY KEY AUTO_INCREMENT,  
19     room_number VARCHAR(10) NOT NULL,  
20     room_type ENUM('VIP', 'Regular') NOT NULL,  
21     price_per_day DECIMAL(10,2) NOT NULL,  
22     room_status ENUM('Available', 'Occupied') DEFAULT 'Available'  
23 );
```

```
25 • INSERT INTO Rooms (room_number, room_type, price_per_day, room_status) VALUES  
26 ('A001', 'VIP', 950000.00, 'Available'),  
27 ('B002', 'Regular', 500000.00, 'Occupied'),  
28 ('C003', 'VIP', 950000.00, 'Available');  
29
```

	room_id	room_number	room_type	price_per_day	room_status	
	1	A001	VIP	950000.00	Available	
	2	B002	Regular	500000.00	Occupied	
	3	C003	VIP	950000.00	Available	
	NULL	NULL	NULL	NULL	NULL	

Admissions (Rawat Inap)

```
31 • CREATE TABLE Admissions (  
32     admission_id INT PRIMARY KEY AUTO_INCREMENT,  
33     patient_id INT,  
34     room_id INT,  
35     admission_date DATE,  
36     discharge_date DATE,  
37     FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
38     FOREIGN KEY (room_id) REFERENCES Rooms(room_id)  
39 );  
40
```

```
41 • INSERT INTO Admissions (patient_id, room_id, admission_date, discharge_date) VALUES  
42 (1, 02, '2025-10-01', '2025-11-05'),  
43 (2, 03, '2025-10-02', '2025-10-30'),  
44 (3, 01, '2025-10-03', NULL);
```

	admission_id	patient_id	room_id	admission_date	discharge_date	
	1	1	2	2025-10-01	2025-11-05	
	2	2	3	2025-10-02	2025-10-30	
	3	3	1	2025-10-03	NULL	
	NULL	NULL	NULL	NULL	NULL	

Doctors (Dokter)

```
30 • ⊖ CREATE TABLE Doctors (  
31     doctor_id INT PRIMARY KEY AUTO_INCREMENT,  
32     name VARCHAR(100),  
33     specialization VARCHAR(100),  
34     phone_number VARCHAR(15),  
35     email VARCHAR(50)  
36 );
```

```
55 • INSERT INTO Doctors (name, specialization, phone_number, email) VALUES  
56     ('Dr. Harits A', 'Penyakit Dalam', '081234567890', 'Harits.A@rs.com'),  
57     ('Dr. M Arrozzaq', 'Kulit', '088990077661', 'M.arrozzaq@rs.com'),  
58     ('Dr. Wahyu Ananda', 'THT', '082783667367', 'wahyu.ananda@rs.com');
```

	doctor_id	name	specialization	phone_number	email
	1	Dr. Harits A	Penyakit Dalam	081234567890	Harits.A@rs.com
	2	Dr. M Arrozzaq	Kulit	088990077661	M.arrozzaq@rs.com
	3	Dr. Wahyu Ananda	THT	082783667367	wahyu.ananda@rs.com
	NULL	NULL	NULL	NULL	NULL

Medical Records (Rekam Medis)

```
38 • ⊖ CREATE TABLE MedicalRecords (  
39     record_id INT PRIMARY KEY AUTO_INCREMENT,  
40     patient_id INT,  
41     doctor_id INT,  
42     diagnosis VARCHAR(255),  
43     treatment TEXT,  
44     record_date DATE,  
45     FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
46     FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
47 );
```

```
72 • INSERT INTO MedicalRecords (patient_id, doctor_id, diagnosis, treatment, record_date) VALUES  
73     (1, 1, 'Ginjal', 'Bedah', '2025-10-01'),  
74     (2, 2, 'Alergi ', 'Pemberian obat salep', '2025-10-02'),  
75     (3, 3, 'Mata', 'Pemberian Obat mata', '2025-10-03');
```

record_id	patient_id	doctor_id	diagnosis	treatment	record_date
4	1	1	Ginjal	Bedah	2025-10-01
5	2	2	Alergi	Pemberian obat salep	2025-10-02
6	3	3	Mata	Pemberian Obat mata	2025-10-03
NULL	NULL	NULL	NULL	NULL	NULL

Payments (Pembayaran)

```

49 CREATE TABLE Payments (
50     payment_id INT PRIMARY KEY AUTO_INCREMENT,
51     admission_id INT,
52     payment_date DATE,
53     total_amount DECIMAL(10,2),
54     payment_status ENUM('Paid', 'Unpaid'),
55     FOREIGN KEY (admission_id) REFERENCES Admissions(admission_id)
56 );

```

```

86 INSERT INTO Payments (admission_id, payment_date, total_amount, payment_status) VALUES
87 (1, '2025-10-03', 1400000.00, 'Paid'),
88 (2, '2025-10-02', 750000.00, 'Paid'),
89 (3, NULL, 0.00, 'Unpaid');

```

payment_id	admission_id	payment_date	total_amount	payment_status
1	1	2025-10-03	1400000.00	Paid
2	2	2025-10-02	750000.00	Paid
3	3	NULL	0.00	Unpaid
NULL	NULL	NULL	NULL	NULL

D. Kesimpulan

Pembuatan database sistem rawat inap, dapat disimpulkan bahwa penerapan konsep basis data relasional mampu membantu dalam pengelolaan informasi rumah sakit secara efisien. Dengan adanya tabel-tabel seperti *Patients*, *Doctors*, *Rooms*, *Admissions*, *MedicalRecords*, dan *Payments*, data dapat saling terhubung dan mudah diakses. Selain itu, penggunaan *foreign key* memastikan konsistensi dan integritas data antar tabel. Melalui percobaan ini, mahasiswa dapat memahami proses perancangan, pembuatan, serta implementasi relasi dalam sistem basis data menggunakan MySQL sebagai platform utama.